

Ian Crawford

Irvine, CA (Open to Relocation) | icrawford8888@gmail.com | 949-738-3879 | icrawfo.github.io

Authorized to work in the U.S. (F-1, CPT/OPT Eligible)

Education

University of California, Irvine

Expected June 2028

B.S. Mechanical Engineering & B.S. Aerospace Engineering (Double Major)

Irvine, CA

GPA: 3.78/4.0 | Dean's List: All Quarters

Relevant Coursework: Computer-Aided Design, Fluid Mechanics, Mechanics of Materials, Vibrations, Engineering Mechanics (Statics & Dynamics), Feedback Control Systems

Experience

Project Management Intern

Jun 2026 – Sep 2026

Fortinet

Vancouver, BC, Canada

- Designed and deployed an internal search tool unifying ~100 spreadsheets used daily by an 8-person team and partner teams, cutting lookups from 10–20 minutes to under 15 seconds; remains in active use.
- Consolidated redundant, siloed tracking spreadsheets across teams into standardized workflows, reducing duplicate data entry and improving cross-team visibility.
- Authored weekly status reports translating project data into actionable summaries for executive stakeholders.

Manufacturing Engineering Intern / Teaching Assistant

Jan 2026 – Present

UCI Undergraduate Machine Shop

Irvine, CA

- Sole undergraduate certified to operate/program UCI's HAAS CNC mill; support ~50+ students weekly across two 12-person course sections plus walk-in shop users.
- Machined components for 15+ faculty and graduate research projects since January 2026, including custom brackets and test fixtures.
- Helped commission the shop's HAAS CNC and establish its CAM training curriculum; optimized toolpaths to cut cycle time on a production wheel-center part for the school's EV team.

Projects

FSAE Chassis Lead (6-Person Subteam)

May 2025 – Present

Anteater Formula Racing (FSAE, 45 Members)

Irvine, CA

- Own design and FEA validation (ANSYS Mechanical) of the current chassis: 1,371 lb-ft/deg torsional rigidity, up 33% from 1,027 lb-ft/deg last season, targeting 58 lb (untabbed, unwelded) vs. 63 lb the prior season.
- Selected 4130 chromoly over 1020 DOM steel, leveraging higher yield strength to cut tube wall thickness while holding structural targets.
- Manufactured and TIG-welded the prior chassis (98 lb w/ tabs, down from 122 lb two seasons earlier) – the program's first one-year build cycle in 20 years – and fabricated the pedal box, suspension, and powertrain components.
- CNC-machined the 6061-aluminum pedal box to ± 0.003 in press-fit tolerances with zero failures; held welded suspension pickup-point accuracy to 1.25 mm average (6 mm worst-case).
- Helped raise the team from 79th overall (no acceleration run) to 51st overall and 10th in acceleration at FSAE Michigan.
- Leading early-stage research (not yet implemented) into bonding carbon fiber to the steel chassis tubing to push torsional rigidity beyond current all-steel limits.

Propulsion & Manufacturing Member

May 2025 – Mar 2026

UCI Rocket Project Solids (30 Members)

Irvine, CA

- Designed manufacturing processes for graphite motor nozzles and 6061-aluminum motor cases/carrier rings; processes remain in use by the team.
- Machined 4 graphite nozzles for a 98mm N-class (~3,000 N avg. thrust) APCP motor: 3 characterization nozzles with center bores held to ± 0.03 in tolerance, and 1 full-scale static-fire nozzle with a ~1 in center bore; propellant characterized at Kn 270 at 1,000 psi chamber pressure.
- Processes established during my tenure remained in use as the team went on to place in the top 20 at IREC.

Technical Skills

CAD/CAM: SolidWorks, Fusion 360 (CAM Certified), Siemens NX (Familiar)

Simulation: ANSYS Mechanical (FEA), SolidWorks Simulation, ANSYS Fluent (CFD, Familiar)

Programming: Python, MATLAB, Simulink (Familiar)

Manufacturing: HAAS CNC (Certified Operator/Programmer), Bridgeport Manual Mill, CNC & Manual Lathe, TIG Welding (steel/stainless to 0.020 in, aluminum to 0.065 in; 150–200 hrs), Plasma Cutting, CNC Routing

Design & Quality: GD&T (ASME Y14.5), Tolerance Stack-Up Analysis, DFM

Certifications: HAAS CNC Operator (UCI); Fusion 360 CAM